

COMPUTER NETWORKS

Ex.No.1	Study of various Network commands used in Linux and Windows: Hands-on practice of various network commands.
Date:	

AIM: - To Study of various Network commands used in Linux and Windows

S.No	Command	Synopsis	Description	Sample Input	Sample Output
1	arp -a	arp -a	Displays the ARP cache, showing the IP addresses and corresponding MAC addresses of devices on the local network.	arp -a	Interface: 192.168.1.10 --- 0x7 Internet Address Physical Address Type 192.168.1.1 00-1a-2b-3c-4d-5e dynamic
2	hostname	hostname	Displays the hostname (computer name) of the local system.	hostname	DESKTOP-PC01
3	ipconfig /all	ipconfig /all	Displays complete TCP/IP configuration, including IP address, subnet mask, gateway, DNS, DHCP, and MAC address.	ipconfig /all	IPv4 Address : 192.168.1.10 Default Gateway : 192.168.1.1 DNS Server : 8.8.8.8 DHCP Enabled : Yes
4	nbtstat -a	nbtstat -a <hostname>	Displays NetBIOS name table of a remote computer for troubleshooting	nbtstat -a DESKTOP-PC01	NetBIOS Remote Machine Name Table Name Type Status DESKTOP-PC01 <00> UNIQUE Registered

S.No	Command	Synopsis	Description	Sample Input	Sample Output
			NetBIOS over TCP/IP.		
5	netstat	netstat	Displays active TCP/IP connections, listening ports, routing tables, and interface statistics.	netstat	Proto Local Address Foreign Address State TCP 192.168.1.10:49762 142.250.190.78:443 ESTABLISHED
6	netstat -r	netstat -r	Displays the IP routing table of the local computer.	netstat -r	IPv4 Route Table Network Destination Netmask Gateway Interface
7	nslookup	nslookup <domain>	Queries the DNS server to obtain the IP address or DNS records of a domain.	nslookup www.google.com	Server: dns.google Address: 8.8.8.8 Name: www.google.com Address: 142.250.190.68
8	pathping	pathping <hostname/IP>	Combines the features of ping and tracert to trace the route and measure packet loss at each hop.	pathping www.google.com	Tracing route to www.google.com Computing statistics... Hop 1 192.168.1.1 Hop 2 10.0.0.1 Packet Loss: 0%
9	ping	ping <hostname/IP>	Tests connectivity between the local computer and another host using ICMP Echo Request and Reply messages.	ping www.facebook.com	Pinging www.facebook.com [157.240.22.35] with 32 bytes of data: Reply from 157.240.22.35: bytes=32 time=25ms TTL=54
10	route	route print	Displays or modifies the IP routing table. It is commonly used to view or configure static routes.	route print	IPv4 Route Table Active Routes: Network Destination Netmask Gateway Interface

Some important Linux networking commands

S.No	Command	Synopsis	Description	Sample Input	Sample Output
1	ip	ip <OPTIONS> <OBJECT> <COMMAND>	Displays and manages IP addresses, routing tables, and network interfaces.	ip address show	2: eth0: <UP,BROADCAST,RUNNING> inet 192.168.1.100/24 brd 192.168.1.255 scope global eth0
2	ip address add	ip address add <IP>/<Prefix> dev <Interface>	Assigns an IP address to a network interface.	ip address add 192.168.1.254/24 dev eth0	Assigned 192.168.1.254/24 to eth0 successfully.
3	ip address del	ip address del <IP>/<Prefix> dev <Interface>	Removes an IP address from a network interface.	ip address del 192.168.1.254/24 dev eth0	Deleted 192.168.1.254/24 from eth0.
4	ip link set up	ip link set <Interface> up	Brings the network interface online.	ip link set eth0 up	Interface eth0 is UP.
5	ip link set down	ip link set <Interface> down	Brings the network interface offline.	ip link set eth0 down	Interface eth0 is DOWN.
6	ip link set promisc on	ip link set <Interface> promisc on	Enables promiscuous mode on the interface.	ip link set eth0 promisc on	Promiscuous mode enabled on eth0.
7	ip route add default	ip route add default via <Gateway> dev <Interface>	Adds the default gateway for all outgoing traffic.	ip route add default via 192.168.1.254 dev eth0	Default route added successfully.
8	ip route add	ip route add <Network> via <Gateway>	Adds a static route through a gateway.	ip route add 192.168.1.0/24 via 192.168.1.254	Route added successfully.

S.No	Command	Synopsis	Description	Sample Input	Sample Output
9	ip route add (Direct Route)	ip route add <Network> dev <Interface>	Adds a direct route through a network interface.	ip route add 192.168.1.0/24 dev eth0	Route added via eth0.
10	ip route delete	ip route delete <Network> via <Gateway>	Deletes a static route.	ip route delete 192.168.1.0/24 via 192.168.1.254	Route deleted successfully.
11	ip route get	ip route get <Destination IP>	Displays the route used to reach a destination.	ip route get 10.10.1.4	10.10.1.4 via 192.168.1.254 dev eth0 src 192.168.1.100
12	ifconfig	ifconfig	Displays and configures network interface parameters (deprecated in modern Linux).	ifconfig	eth0: flags=4163<UP,BROADCAST,RUNNING> inet 192.168.1.100 netmask 255.255.255.0
13	mtr	mtr <hostname/IP>	Combines Ping and Traceroute to analyze network paths.	mtr google.com	HOST: localhost Loss% Snt Last Avg Best Wrst StDev
14	mtr -g	mtr -g <hostname>	Displays numeric IP addresses instead of hostnames.	mtr -g google.com	192.168.1.1 → 142.250.190.78
15	mtr -b	mtr -b <hostname>	Displays both hostnames and IP addresses.	mtr -b google.com	router.local (192.168.1.1) google.com (142.250.190.78)
16	mtr -c	mtr -c <count> <hostname>	Sends a specified number of probes.	mtr -c 10 google.com	10 probes completed successfully.
17	tcpdump -D	tcpdump -D	Lists all available	tcpdump -D	1.eth0 2.lo 3.wlan0

S.No	Command	Synopsis	Description	Sample Input	Sample Output
			network interfaces.		
18	tcpdump	tcpdump -i <Interface>	Captures packets on a network interface.	tcpdump -i eth0	Listening on eth0...
19	tcpdump -c	tcpdump -i <Interface> -c <count>	Captures a specified number of packets.	tcpdump -i eth0 -c 10	10 packets captured.
20	tcpdump host	tcpdump -i <Interface> host <IP>	Captures traffic to and from a specific host.	tcpdump -i eth0 host 8.8.8.8	Captured packets to/from 8.8.8.8.
21	tcpdump src host	tcpdump -i <Interface> src host <IP>	Captures packets originating from a specific host.	tcpdump -i eth0 src host 8.8.8.8	Source packets from 8.8.8.8 displayed.
22	tcpdump dst host	tcpdump -i <Interface> dst host <IP>	Captures packets destined for a specific host.	tcpdump -i eth0 dst host 8.8.8.8	Destination packets to 8.8.8.8 displayed.
23	tcpdump net	tcpdump -i <Interface> net <Network>/<Prefix>	Captures traffic from a specified network.	tcpdump -i eth0 net 10.1.0.0/24	Packets from network 10.1.0.0/24 displayed.
24	tcpdump port	tcpdump -i <Interface> port <Port>	Captures traffic on a specific port.	tcpdump -i eth0 port 53	DNS packets captured.
25	tcpdump host and port	tcpdump -i <Interface> host <IP> and port <Port>	Captures traffic for a specific host and port.	tcpdump -i eth0 host 8.8.8.8 and port 53	DNS traffic with host 8.8.8.8 displayed.
26	tcpdump port not	tcpdump -i <Interface> port not <Port1> and not <Port2>	Captures all traffic except the specified ports.	tcpdump -i eth0 port not 53 and not 25	Traffic excluding ports 53 and 25 captured.
27	ping	ping <hostname/IP>	Tests network connectivity using ICMP	ping google.com	64 bytes from google.com: icmp_seq=1 ttl=117 time=10.5 ms

S.No	Command	Synopsis	Description	Sample Input	Sample Output
			Echo Requests and Echo Replies.		
28	ping -c	ping -c <count> <hostname>	Sends a specified number of ICMP Echo Requests.	ping -c 10 google.com	10 packets transmitted, 10 received, 0% packet loss.

Configuring an Ethernet connection by using nmcli:

Step No.	Task	Command	Expected Result
1	List all Network Manager connection profiles	nmcli connection show	Displays all available network connection profiles with their Name, UUID, Type, and Device .
2	Create a new Ethernet connection profile It is specifically an nmcli connection management command. • nmcli → NetworkManager Command Line Interface • connection → Manages network connection profiles. • add → Creates a new connection profile.	nmcli connection add con-name Internal-LAN ifname enp1s0 type ethernet	Creates a new Ethernet connection profile named Internal-LAN .
3	Rename an existing connection profile (Optional)	nmcli connection modify "Wired connection 1" connection.id Internal-LAN	Renames the existing connection profile to Internal-LAN .

Step No.	Task	Command	Expected Result
4	Display the current settings of the connection profile	<code>nmcli connection show "Internal-LAN"</code>	Displays interface name, autoconnect status, IPv4 and IPv6 configuration details.
5	Configure IPv4 using DHCP	<code>nmcli connection modify "Internal-LAN" ipv4.method auto</code>	Configures the interface to obtain the IPv4 address automatically using DHCP.
6	Configure a static IPv4 address	<code>nmcli connection modify "Internal-LAN" ipv4.method manual ipv4.addresses 192.0.2.1/24 ipv4.gateway 192.0.2.254 ipv4.dns 192.0.2.200 ipv4.dns-search example.com</code>	Assigns a static IPv4 address, gateway, DNS server, and DNS search domain.
7	Configure IPv6 using SLAAC	<code>nmcli connection modify "Internal-LAN" ipv6.method auto</code>	Enables automatic IPv6 configuration using Stateless Address Autoconfiguration (SLAAC).
8	Configure a static IPv6 address	<code>nmcli connection modify "Internal-LAN" ipv6.method manual ipv6.addresses 2001:db8:1::fffe/64 ipv6.gateway 2001:db8:1::ffee ipv6.dns 2001:db8:1::ffbb ipv6.dns-search example.com</code>	Assigns a static IPv6 address, gateway, DNS server, and search domain.
9	Activate the connection profile	<code>nmcli connection up Internal-LAN</code>	Activates the Internal-LAN network connection profile.
10	Verify the IP address configuration	<code>ip address show enp1s0</code>	Displays the assigned IPv4 and IPv6 addresses of the enp1s0 network interface.
11	Verify the IPv4 default gateway	<code>ip route show default</code>	Displays the configured IPv4 default gateway.
12	Verify the IPv6 default gateway	<code>ip -6 route show default</code>	Displays the configured IPv6 default gateway.

Step No.	Task	Command	Expected Result
13	Verify the DNS configuration	cat /etc/resolv.conf	Displays the configured DNS servers and DNS search domain.
14	Test network connectivity	ping <host-name-or-IP-address>	Sends ICMP Echo Requests to verify connectivity with the specified host.

Sample Verification Outputs

Verification Command	Sample Output
ip address show enp1s0	inet 192.0.2.1/24 inet6 2001:db8:1::fffe/64
ip route show default	default via 192.0.2.254 dev enp1s0 proto static
ip -6 route show default	default via 2001:db8:1::fffe dev enp1s0 proto static
cat /etc/resolv.conf	search example.com nameserver 192.0.2.200 nameserver 2001:db8:1::ffbb
ping google.com	64 bytes from google.com: icmp_seq=1 ttl=117 time=11.2 ms

RESULT:

```
File Edit View Search Terminal Help

My traceroute [v0.95]
fedora (10.0.2.15) -> google.com (142.251.221.206) 2026-07-25T09:05:57+0530
Keys: Help Display mode Restart statistics Order of fields quit

Packets
Pings
Host Loss% Snt Last Avg Best Wrst StDev
1. pnmaaa-ba-in-f14.1e100.net 0.0% 4 6.4 6.9 6.4 7.3 0.5
```

```
File Edit View Search Terminal Help

My traceroute [v0.95]
fedora (10.0.2.15) -> google.com (142.251.220.110) 2026-07-25T09:07:39+0530
Keys: Help Display mode Restart statistics Order of fields quit

          Packets
Host      Loss%  Snt  Last  Avg  Best  Wrst StDev
1. 142.251.220.110 (142.251.220.1 0.0%   4  13.0 18.0 13.0 30.0  8.0
```

```
student@fedora:~$ tcpdump -D
1.enp0s3 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.usbmon2 (Raw USB traffic, bus number 2)
6.usbmon1 (Raw USB traffic, bus number 1)
7.usbmon0 (Raw USB traffic, all USB buses) [none]
8.nflog (Linux netfilter log (NFLOG) interface) [none]
9.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
student@fedora:~$
```

```
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student@fedora:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac prefixlen 64 scopeid 0x0<
global>
    inet6 fe80::55f0:9771:cb55:7bf3 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:2d:91:bf txqueuelen 1000 (Ethernet)
    RX packets 138 bytes 29665 (28.9 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 168 bytes 17540 (17.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 20 bytes 2130 (2.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 20 bytes 2130 (2.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

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```
student@fedora:~$ arp -a
```

```
_gateway (10.0.2.2) at 52:55:0a:00:02:02 [ether] on enp0s3
```

```
student@fedora:~$ hostname
```

```
fedora
```

```
student@fedora:~$ route print
```

```
Usage: route [-nNvee] [-FC] [<AF>]          List kernel routing tables
```

```
       route [-v] [-FC] {add|del|flush} ...  Modify routing table for AF.
```

```
       route {-h|--help} [<AF>]             Detailed usage syntax for specifi  
ed AF.
```

```
       route {-V|--version}                 Display version/author and exit.
```

```
       -v, --verbose                        be verbose
```

```
       -n, --numeric                       don't resolve names
```

```
       -e, --extend                        display other/more information
```

```
       -F, --fib                           display Forwarding Information Base (default)
```

```
       -C, --cache                         display routing cache instead of FIB
```

```
<AF>=Use -4, -6, '-A <af>' or '--<af>'; default: inet
```

```
List of possible address families (which support routing):
```

```
inet (DARPA Internet) inet6 (IPv6) ax25 (AMPR AX.25)
```

```
netrom (AMPR NET/ROM) ipx (Novell IPX) ddp (Appletalk DDP)
```

```
x25 (CCITT X.25)
```

File Edit View Search Terminal Help

```
student@fedora:~127$ ping www.facebook.com
```

```
PING star-mini.c10r.facebook.com (57.144.54.1) 56(84) bytes of data.
```

```
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=1 ttl=255 time=15.9 ms
```

```
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=2 ttl=255 time=22.2 ms
```

```
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=3 ttl=255 time=21.4 ms
```

```
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=4 ttl=255 time=12.8 ms
```

```
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=5 ttl=255 time=13.4 ms
```

```
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=6 ttl=255 time=8.70 ms
```

```
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=7 ttl=255 time=7.51 ms
```

```
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=8 ttl=255 time=7.48 ms
```

```
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=9 ttl=255 time=10.1 ms
```

```
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=10 ttl=255 time=9.73 ms
```

```
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=11 ttl=255 time=11.0 ms
```

```

File Edit View Search Terminal Help
student@fedora:~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 fedora:bootpc          _gateway:bootps        ESTABLISHED

Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State       I-Node  Path
unix    3      [ ]          STREAM     CONNECTED   19174    /run/systemd/journal/stdout
unix    3      [ ]          STREAM     CONNECTED   17819    /run/systemd/journal/stdout
unix    3      [ ]          STREAM     CONNECTED   23930
unix    3      [ ]          STREAM     CONNECTED   22896
unix    3      [ ]          STREAM     CONNECTED   22825    /run/systemd/journal/stdout
unix    3      [ ]          STREAM     CONNECTED   22567    /run/user/1000/bus
unix    3      [ ]          STREAM     CONNECTED   19354    /run/dbus/system_bus_socket
unix    3      [ ]          STREAM     CONNECTED   22624    /run/dbus/system_bus_socket
unix    3      [ ]          STREAM     CONNECTED   22817    /run/user/1000/bus
unix    3      [ ]          STREAM     CONNECTED   22591    /run/dbus/system_bus_socket

```

```

student@fedora:~$ netstat -r
Kernel IP routing table
Destination      Gateway         Genmask         Flags   MSS Window  irtt Iface
default          _gateway       0.0.0.0         UG      0 0        0 enp0s3
10.0.2.0         0.0.0.0        255.255.255.0   U       0 0        0 enp0s3

student@fedora:~$ nslookup google.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.77.110
Name:   google.com
Address: 2404:6800:4007:802::200e

student@fedora:~$

```



Command Prompt



C:\Users\DELL>nbtstat -a HDC0124105

Ethernet 2:

Node IpAddress: [192.168.56.1] Scope Id: []

Host not found.

Ethernet:

Node IpAddress: [0.0.0.0] Scope Id: []

Host not found.

WiFi:

Node IpAddress: [172.16.75.45] Scope Id: []

Host not found.

Local Area Connection* 9:

Node IpAddress: [0.0.0.0] Scope Id: []

Host not found.

Local Area Connection* 10:

Node IpAddress: [0.0.0.0] Scope Id: []

Host not found.

Bluetooth Network Connection:

Node IpAddress: [0.0.0.0] Scope Id: []

Host not found.

C:\Users\DELL>

```
C:\Users\DELL>pathping HDC0124105
```

```
Tracing route to HDC0124105 [172.16.75.45]  
over a maximum of 30 hops:
```

```
 0  HDC0124105 [172.16.75.45]  
 1  HDC0124105 [172.16.75.45]
```

```
Computing statistics for 25 seconds...
```

Hop	RTT	Source to Here Lost/Sent = Pct	This Node/Link Lost/Sent = Pct	Address
0				HDC0124105 [172.16.75.45]
			0/ 100 = 0%	
1	0ms	0/ 100 = 0%	0/ 100 = 0%	HDC0124105 [172.16.75.45]

```
Trace complete.
```

```
C:\Users\DELL>
```

```
student@fedora:~127$ ip address show
```

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
```

```
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
```

```
    inet 127.0.0.1/8 scope host lo
```

```
        valid_lft forever preferred_lft forever
```

```
    inet6 ::1/128 scope host noprefixroute
```

```
        valid_lft forever preferred_lft forever
```

```
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
```

```
    link/ether 08:00:27:2d:91:bf brd ff:ff:ff:ff:ff:ff
```

```
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
```

```
        valid_lft 83835sec preferred_lft 83835sec
```

```
    inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac/64 scope global dynamic noprefixroute
```

```
        valid_lft 86135sec preferred_lft 14135sec
```

```
    inet6 fe80::55f0:9771:cb55:7bf3/64 scope link noprefixroute
```

```
        valid_lft forever preferred_lft forever
```

```
student@fedora:~$
```

File Edit View Search Terminal Help

```
student@fedora:~$ mtr -g google.com
```

```
mtr: invalid option -- 'g'
```

Usage:

```
mtr [options] hostname
```

```
-F, --filename FILE      read hostname(s) from a file
-4                        use IPv4 only
-6                        use IPv6 only
-u, --udp                use UDP instead of ICMP echo
-T, --tcp                use TCP instead of ICMP echo
-I, --interface NAME     use named network interface
-a, --address ADDRESS    bind the outgoing socket to ADDRESS
-f, --first-ttl NUMBER   set what TTL to start
-m, --max-ttl NUMBER     maximum number of hops
-U, --max-unknown NUMBER maximum unknown host
-P, --port PORT          target port number for TCP, SCTP, or UDP
-L, --localport LOCALPORT source port number for UDP
-s, --psize PACKETSIZE   set the packet size used for probing
-B, --bitpattern NUMBER  set bit pattern to use in payload
-i, --interval SECONDS   ICMP echo request interval
-G, --gracetime SECONDS  number of seconds to wait for responses
-Q, --tos NUMBER         type of service field in IP header
-e, --mpls               display information from ICMP extensions
```

File Edit View Search Terminal Help

My traceroute [v0.95]

```
fedora (10.0.2.15) -> google.com (142.250.182.78) 2026-07-25T09:09:18+0530
```

```
Keys: Help  Display mode  Restart statistics  Order of fields  quit
```

Host	Packets		Pings				
	Loss%	Snt	Last	Avg	Best	Wrst	StDev
1. lcmaaa-ax-in-f14.1e100.net	0.0%	3	8.1	8.8	8.1	9.7	0.8

```
student@fedora:~$ nmcli connection show
```

NAME	UUID	TYPE	DEVICE
Wired connection 1	86ead386-e6ab-359e-9d82-71491148899f	ethernet	enp0s3
lo	d3bc006d-3b51-4d5e-ba62-b097234a30ab	loopback	lo

```
student@fedora:~$
```

```
student@fedora:~$ cat /etc/resolv.conf
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-resolved(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs should typically not access this file directly, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.
```